AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 36 · STUDENT LESSON

Neural network experiment defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Can a revised model reuse inspected final rows as new independent evidence?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Defend a network experiment using traceable evidence. Try the recall before opening notes.

Your learning target

I can defend a network experiment using traceable **evidence**.

I can separate selection, final evaluation and revision claims.

I can answer a challenge with a calculation and a bounded conclusion.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone **limitation**s and revision. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Defend a network experiment using traceable **evidence**

Separate selection, final evaluation and revision claims

Answer a challenge with a calculation and a bounded conclusion

Retrieve what you already know

> Can a revised model reuse inspected final rows as new independent **evidence**?

My starting explanation

This week's end goal

Explain neural network experiment **defense** with a traceable toy-network case and a bounded conclusion.

Notice the evidence

> Investigate the supplied Neural network experiment defense case. Separate observations, given facts and assumptions before choosing an explanation.

A **defense** is not a sales pitch. Assemble the original question and criterion, data roles, baseline, model equation, settings, raw checkpoints, validation selection, frozen final **prediction**s, **limitation**s and revision history.

Someone reading the record should be able to recompute a reported MSE and see which decisions each dataset informed.

For this capstone, two rates were compared under matched conditions. Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 for rate 0.1.

The frozen candidate then achieved final MSE 5 versus baseline 20.25 on four constructed cases. These are different comparisons and must not be collapsed into “the network was 100% accurate”.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Neural network experiment defense. Connect each labelled input, deciding step and result.

Stage	Evidence	Supported statement
Validation selection	Two positive rows; MSE 0 versus 0.78125	Selected rate 0.2 by written rule
Frozen final comparison	Four rows; MSE 5 versus 20.25	Lower MSE than initial-weight baseline on these rows
Condition check	Two negative rows; candidate MSE 10	Nonnegative output fails to represent negative targets
Proposed revision	Signed output on revealed cases	A possible representational repair, not new final evidence

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Question and settings	Replay calculations	Procedure can be checked
Frozen final predictions	Compare same four rows	Bounded baseline improvement
Known failures and changes	Preserve old results	Revision still needs evaluation

Every claim has an evidence source and a limit. A complete record is not a substitute for broader data.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Neural network experiment defense. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

State the validation selection and frozen final comparison using their correct metrics and scopes.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Change the second **input** from 1 to 2.

Predicted result

Prediction: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Why the result follows

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Neural network experiment defense to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Defense	An explanation of methods and claims supported by inspectable evidence
Evidence chain	The connection from inputs and procedure through raw outputs to a reported claim
Limitation	A condition restricting what evidence can establish
Reproduction	Repeating the documented procedure to check reported outputs

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Neural network experiment defense, predict a step, then compare with its explanation.

Worked explanation · Neural network experiment defense

Challenge: “Your validation error was zero, so why did the final test fail on negative targets?” Response: validation contained only positive **input**s; the **output** max(0,a*x) cannot be negative.

The final negative cases expose a coverage and architecture **limitation**. The overall criterion still passed, but that does not remove the failure.

Worked explanation · Neural network experiment defense

Challenge: “Is a neural network better than linear regression?” Response: the supplied baseline was a fixed initial weight, not a fitted line. The **evidence** establishes improvement relative to that baseline only.

A fitted signed line is a relevant additional comparison and needs a documented, fair evaluation. Challenge: “Was the candidate faster?” Response: no timing **evidence** was collected, so speed is unmeasured.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Neural network experiment defense

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

The facts we will use

Supplied facts

Inputs are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

AI ACADEMY / AI-5/36

One toy ReLU neuron: not a probability

Input 2 x weight 0.5 = 1

Input 1 x weight -1 = -1

Add the products and bias 0.2: total = 0.2

ReLU: use max(0, total) -> output 0.2

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/36

Neural network experiment defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply inputs

2 x 0.5=1 and 1 x (-1)=-1.





2. Add weighted terms

Pre-activation z is 1-1+0.2=0.2.





3. Apply ReLU

max(0,0.2)=0.2; this activation is not a confidence percentage.

CHANGE ONE THING

Change the second input from 1 to 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Multiply inputs	2×0.5=1 and 1×(-1)=-1.
2. Add weighted terms	Pre-activation z is 1-1+0.2=0.2.
3. Apply ReLU	max(0,0.2)=0.2; this activation is not a confidence percentage.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

What this example does not establish

Keep the stated limit

This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

Keep the stated limit

Read the labelled picture: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

Keep the stated limit

Change one thing: Change the second **input** from 1 to 2. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Neural network experiment defense claim and explain a repair using the supplied evidence.

Check the claim against the facts

Supported case: another experiment reports baseline MSE 8 and candidate MSE 2 on four untouched final rows after freezing the candidate. The reduction is 6 MSE units, or 75% of the baseline MSE.

That percentage is a relative reduction in MSE, not 75% classification accuracy. It cannot establish deployment performance.

Guided: calculate the absolute and relative MSE reduction, then write a sentence preserving the metric and four-row scope.

Locate the earliest unsupported step

A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Neural network experiment defense with support, then explain the deciding step in your own words.

Name eight items needed to check the capstone from question through final result.

Guided: calculate the absolute and relative MSE reduction, then write a sentence preserving the metric and four-row scope.

Explain your prediction to a partner before comparing results.

Practise with less help: Defend the result

Use workbook W2 for the connected example “Neural network experiment **defense**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Neural network experiment **defense**. Proposed explanation: “A polished demonstration is an evaluation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Neural network experiment defense. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

State the validation selection and frozen final comparison using their correct metrics and scopes.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Neural network experiment defense. Explain why each step is justified.

Explain → apply → challenge

Name eight items needed to check the capstone from question through final result.

> A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?

Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0.

The **written rule** selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Use a short presentation: question; method and controls; selection; final result; failures; revision; next question. State the small constructed sample, absent real-world coverage, limited baseline and unmeasured resource use.

A useful next question is whether a signed-**output** procedure improves on a fitted-line baseline under newly reserved evaluation conditions.

Passing this capstone means explaining the **evidence** correctly, including an unsuccessful or limited result.

> **Exit: correct "our network is universally reliable, 100% accurate, faster than every baseline and repaired on the final test". Give measured claims and identify the missing evidence.**

Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

> **My limited claim and a useful next check**

Study a complete case

> Revisit the complete Neural network experiment defense case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Question and settings	Replay calculations	Procedure can be checked
Frozen final predictions	Compare same four rows	Bounded baseline improvement
Known failures and changes	Preserve old results	Revision still needs evaluation

Every claim has an evidence source and a limit. A complete record is not a substitute for broader data.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Defend a new record independently

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
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> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Neural network experiment defense, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

By the end, I can	How I will show it
Defend a network experiment using traceable evidence	Explain the deciding step, show a trace or test, and state one limit.
Separate selection, final evaluation and revision claims	Explain the deciding step, show a trace or test, and state one limit.
Answer a challenge with a calculation and a bounded conclusion	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Neural network experiment defense without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Defend a network experiment using traceable **evidence**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? max(0,0.2)=0.2; this activation is not a confidence percentage.

1. Add weighted terms

2. Apply ReLU

3. Multiply **input**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply ReLU max(0,0.2)=0.2; this activation is not a confidence percentage.

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the second **input** from 1 to 2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. z becomes 1-2+0.2=-0.8 and ReLU returns 0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: z becomes 1-2+0.2=-0.8 and ReLU returns 0. The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Repair a missing building block

Review Capstone **limitation**s and revision (ai-5/35). Goal: Evaluate a frozen candidate and baseline on final rows. Short practice: Calculate the slice rates and explain the claim revision.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?. Return to this week and explain what you changed.

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

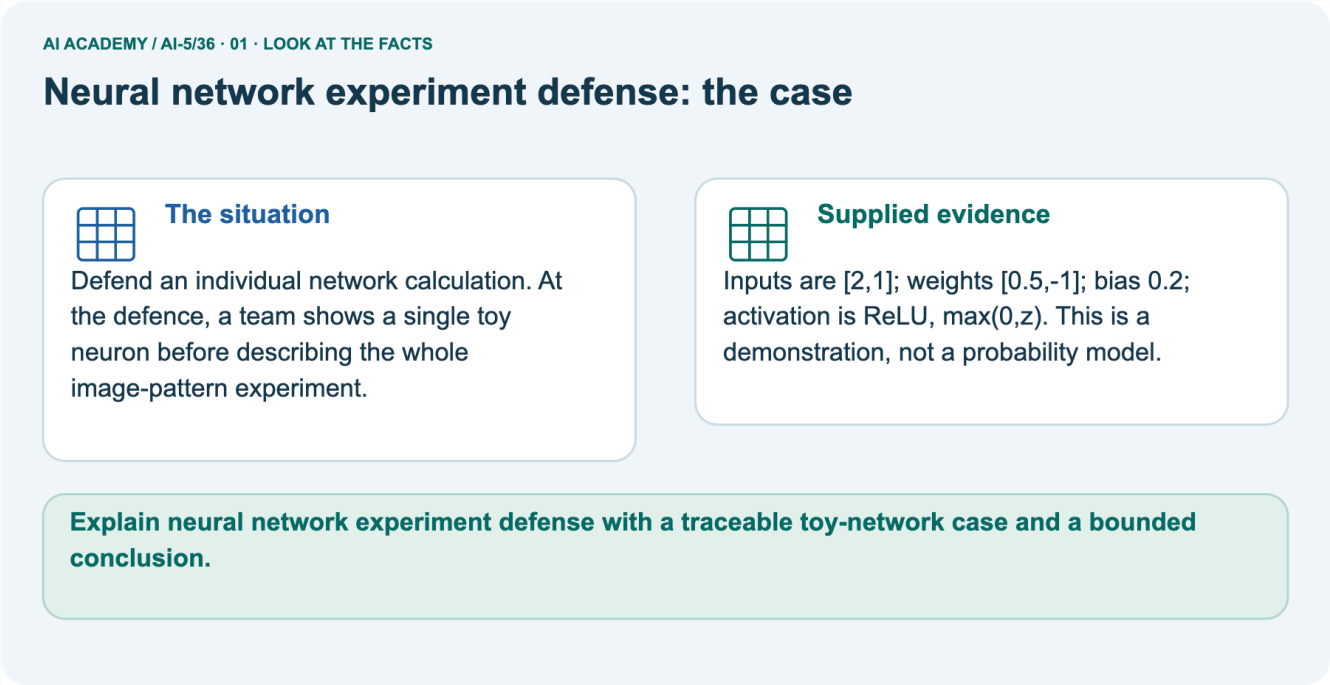
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Neural network experiment defense graphic using its labels, deciding evidence and relevant limitation.

01. Neural network experiment defense: the case



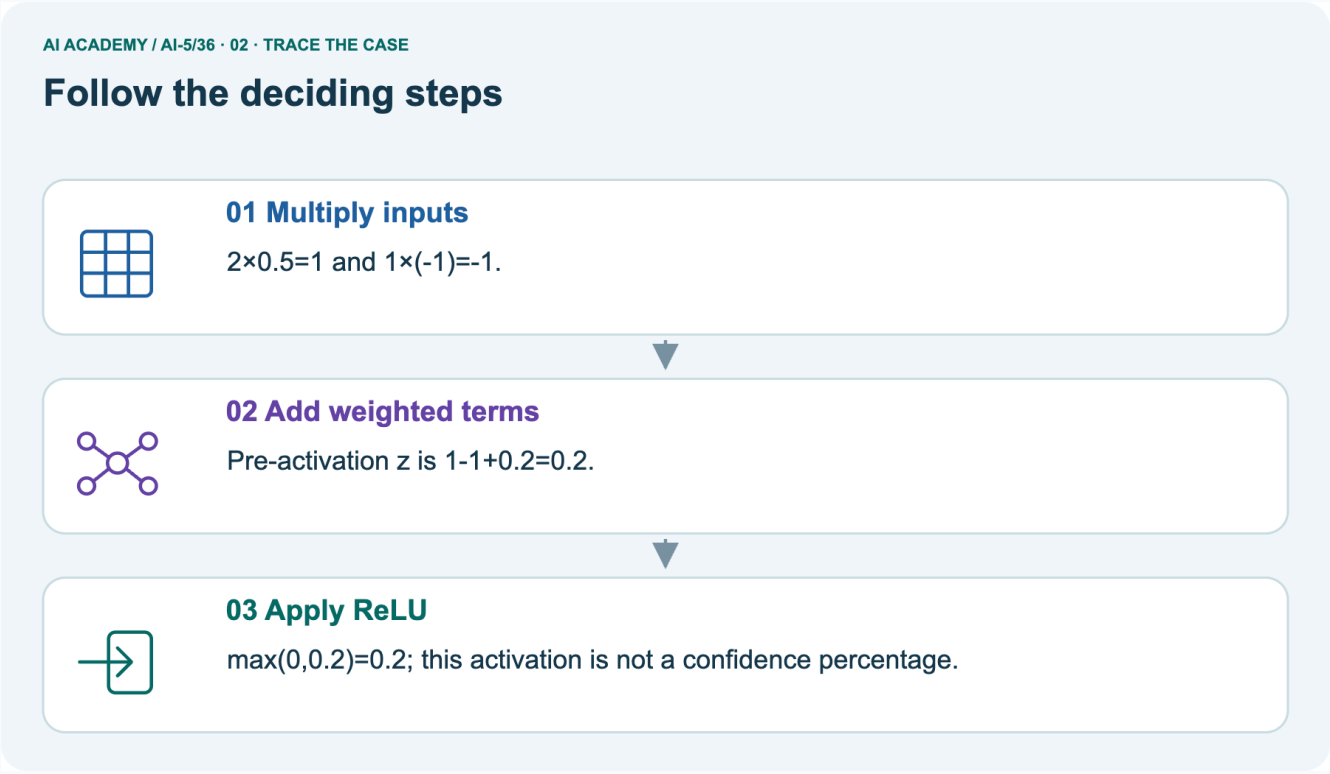
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Supplied facts: **Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

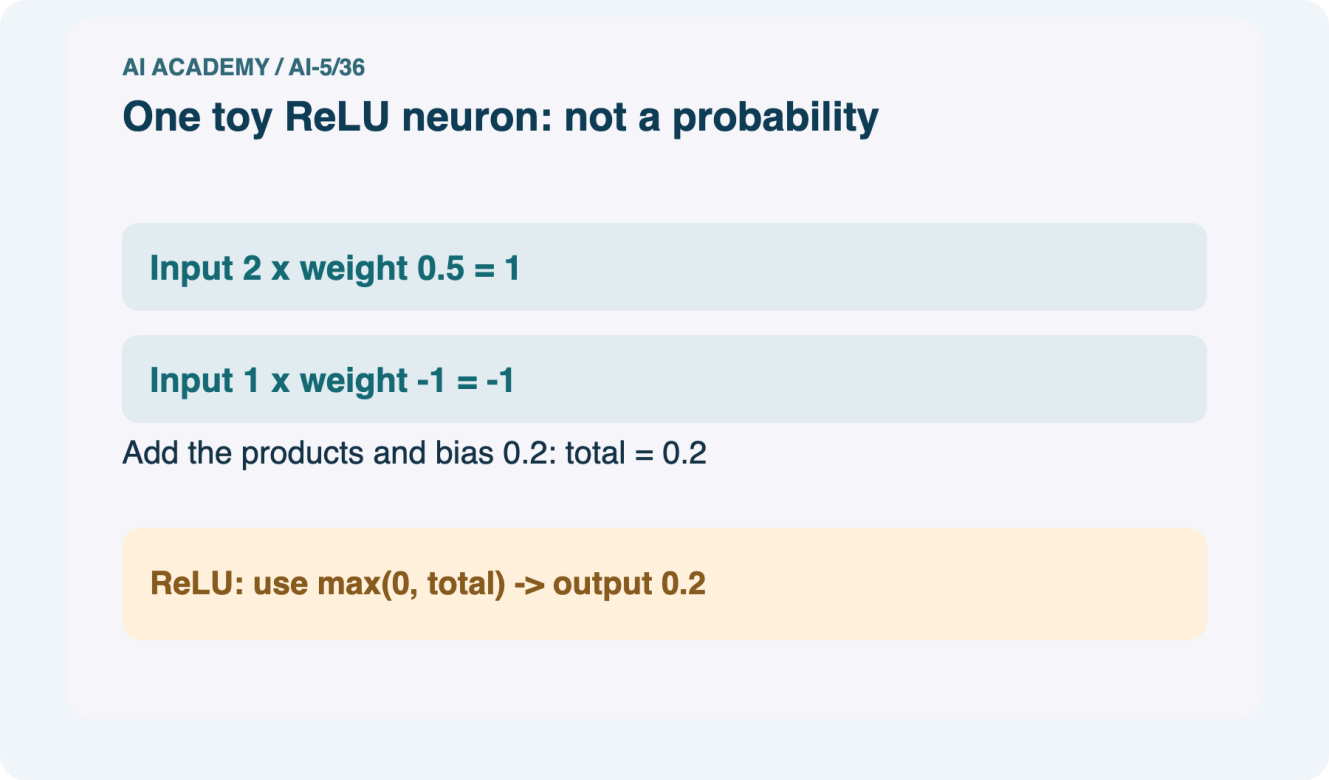


Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply **input**s: 2×0.5=1 and 1×(-1)=-1. → Add weighted terms: Pre-activation z is 1-1+0.2=0.2. → Apply ReLU: max(0,0.2)=0.2; this activation is not a confidence percentage.

> Point to each numbered stage and explain which recorded **input** it uses.

03. One toy ReLU neuron: not a probability

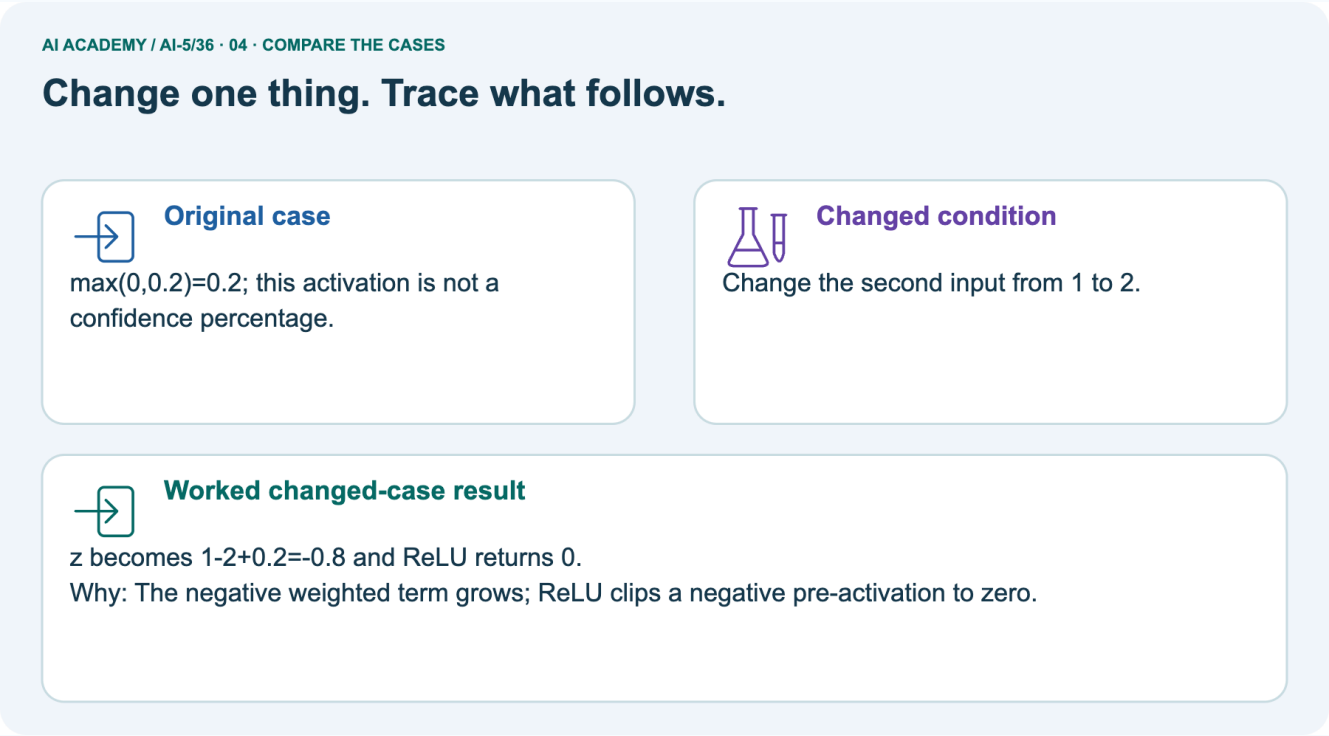


Follow the labelled picture with the matching explanation. Open the vector diagram

One toy ReLU neuron: not a probability. **Input**s: [2, 1]; weights: [0.5, -1]; bias: 0.2; activation: relu; **output**: 0.2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



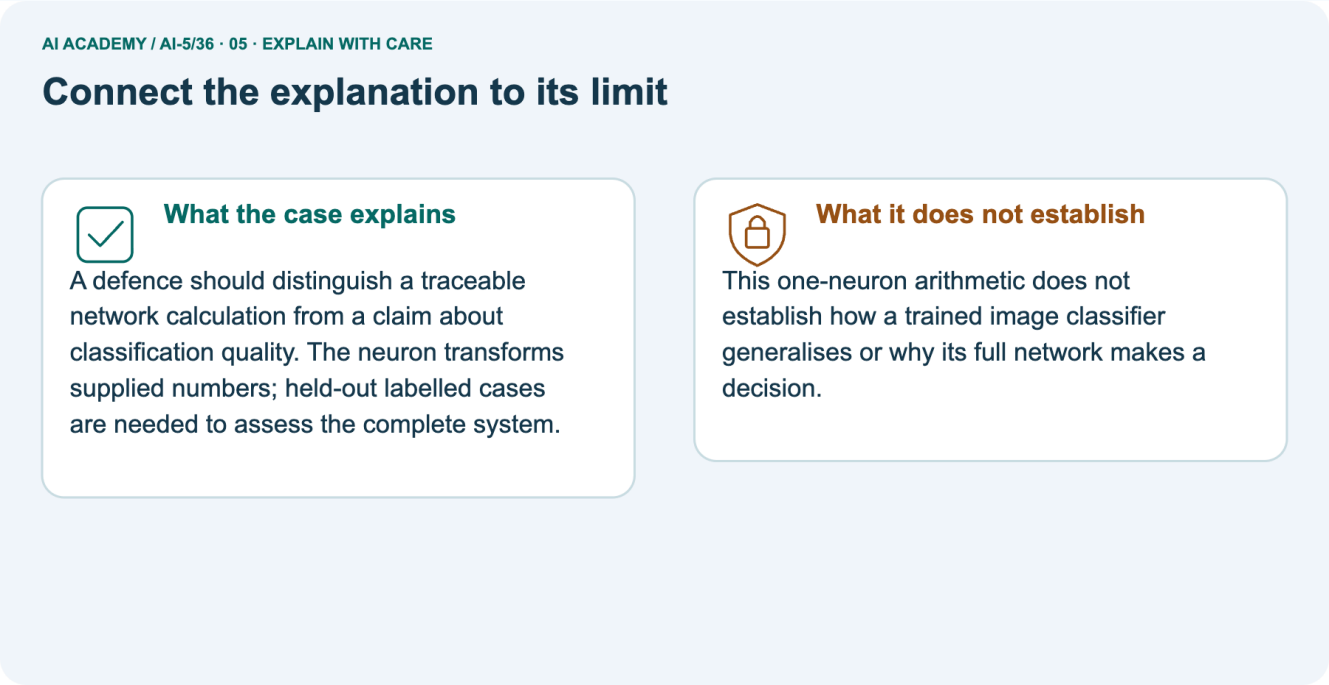
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: max(0,0.2)=0.2; this activation is not a confidence percentage. Changed condition: Change the second **input** from 1 to 2. Expected result: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



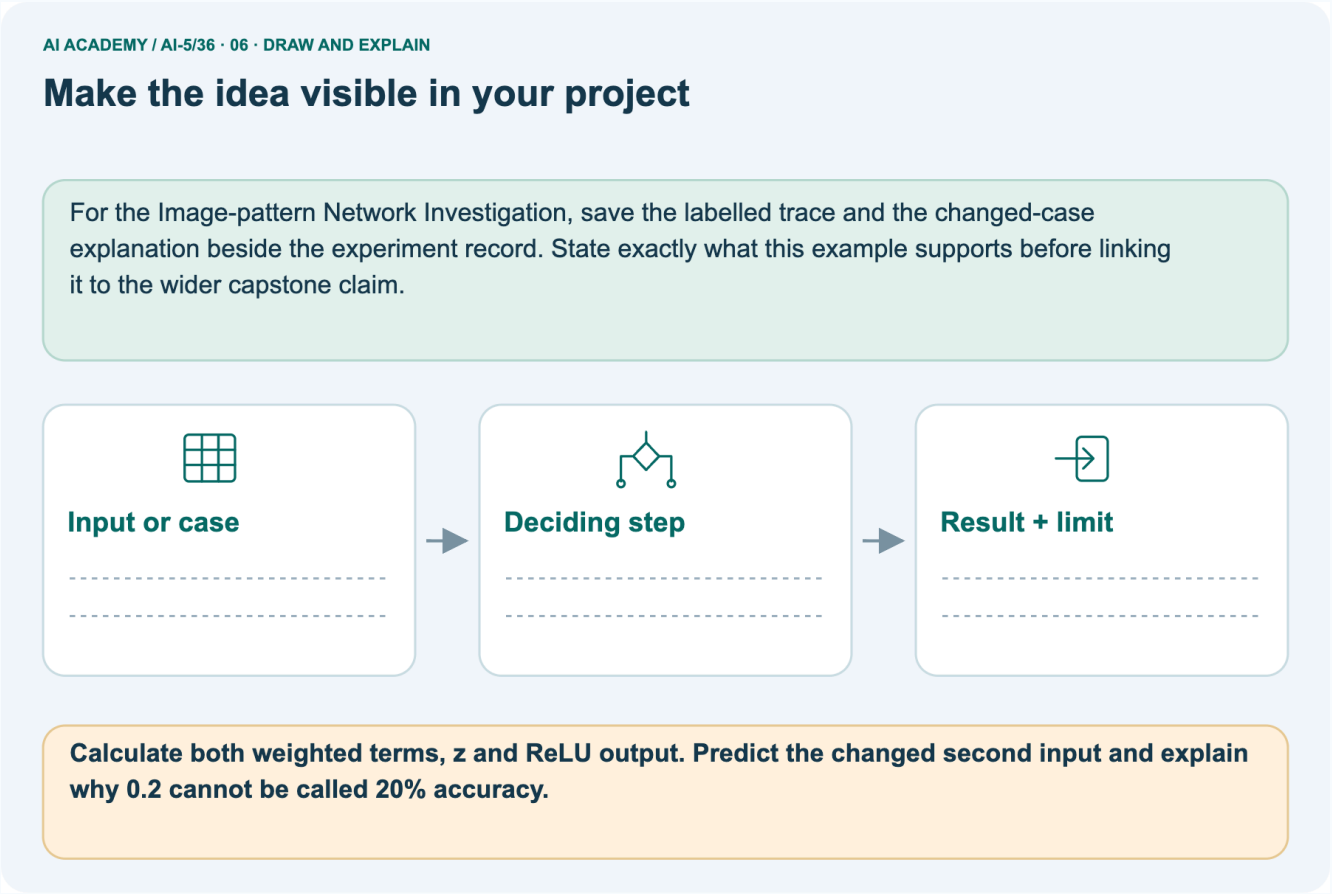
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Limit: This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.